

Hongji Pu

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RESEARCH FOCUS

My research focuses on **self-evolving language agents**, **graph world models**, and **trustworthy LLM systems**. I study how agents can learn from experience, external knowledge, and verified feedback to improve their memory, reusable skills, and decision-making, and how world models can support algorithm design on complex networks. Across these directions, I develop methods and evaluations for systems that continually adapt to new tasks and environments, with an emphasis on the reliability of their behavior and improvements.

EDUCATION

University of Illinois, Urbana-Champaign **Aug. 2025 - Dec. 2026**
M.S. in Financial Engineering GPA: 3.54/4.00 (upon Second Semester) *Champaign, IL, USA*

Central University of Finance and Economics **Aug. 2021 - Jun. 2025**
B.S. in Finance GPA: 3.84/4.00 (Class Rank 1/39) *Beijing, CN*

PUBLICATIONS

Manuscripts and Preprints

* denotes equal contribution

- 2026 **COUNTERMEM: World-Model-Verified Counterfactual Memory for Language Agents**
Hongji Pu, Ruixiang Tong, Yongfeng Zhang
[Paper](#) Under review at ICLR 2027
- 2026 **Network World Models as Environments for Algorithm Design on Complex Networks**
Hongji Pu*, Rishab Alagharu*, Zeeshan Memon, Xinyuan Song, Yuntong Hu, Liang Zhao
[Paper](#) [Code](#) Under review at ICLR 2027
- 2026 **PAPER2LLM++: Continual Self-Evolution of Language Models From Research Papers**
Hongji Pu, Yilun Zhao, Wenpeng Yin
[Paper](#) Under review at TACL 2027.10
- 2026 **Launder-Bench: Can Repository-Level Agentic Code Laundering Be Detected?**
Yuntong Hu, Hongji Pu, Zimo Li, Ruomeng Li, Xiaotong Liu, Ziyuan Qin, Qiyue Li, Liang Zhao
[Code](#) Under review at ICLR 2027
- 2026 **RubricWorld: World Model-Guided Rubric Optimization for LLM Graders**
Hongji Pu, Hang Li, Yucheng Chu, Jiliang Tang
Under review at AAAI 2026
- 2026 **SkillOps: Managing LLM Agent Skill Libraries as Self-Maintaining Software Ecosystems**
Hongji Pu*, Xinyuan Song*, Liang Zhao
[Paper](#) [Code](#) Under review at ICLR 2027
- 2026 **BeliefRAG: Making Adaptive RAG State-Aware Under Evolving Evidence**
Hongji Pu, Leyang Zhou
[Paper](#) [Code](#) Under review at TACL

- 2026 **Beyond Forecasting: Recasting Volatility Control as a Routing Problem**
 Hongji Pu, Leyang Zhou
[Paper](#) [Code](#) Under review at ACM ICAIF 2026
- 2026 **RouteProfile: Graph-Based Profiling for cold-start LLM Routing**
 Hongji Pu*, Jingjun Xu*, Tao Feng*, Haozhen Zhang, Jiaxuan You, Ge Liu
[Paper](#) [Code](#) Preprint

RESEARCH EXPERIENCE

CounterMem: World-Model-Verified Counterfactual Memory for Language Agents **Apr. 2026 – Present**
 Rutgers University, Web Intelligent Systems and Engineering Lab Prof. Yongfeng Zhang, Piscataway, NJ

- ◆ Developed **CounterMem**, an RL-based framework for constructing and reusing verified counterfactual memory in language agents. The method generates local alternatives to failed actions, verifies their outcomes with executable world models, stores action contrasts with evidence and reuse conditions, and trains a DQN selector to decide when retrieved memory should guide a new task.
- ◆ Evaluated on **12 benchmarks across six domains**. CounterMem improved ReAct and Reflexion in **24/24 benchmark-agent comparisons** with gpt-oss-120b, averaging **+12.6 pp**; across two backbones, four-domain gains reached **8.2–20.4 pp** while task-run token usage decreased by **7.7–42.0%**.

RubricGuard: Improving LLM Graders Without Breaking Correct Judgments **Mar. 2026 – Present**
 Michigan State University, Data Science and Engineering Lab (DSE Lab) Prof. Jiliang Tang, East Lansing, MI

- ◆ Developed **RubricGuard**, a Route–Generate–Audit framework for refining LLM grading rubrics without retraining the underlying model. The framework localizes failures to specific score boundaries, generates targeted rubric patches, and audits updates for target improvement and protected-boundary regressions to limit scoring-standard drift.
- ◆ Evaluated on **ASAP 2.0, EIR, and ASAP-SAS across three LLM backbones**, achieving the best performance in **20 of 27 metric–dataset–backbone settings**. Boundary-level ablations showed that retaining the audit component reduced protected-case breakage from **15.7% to 3.6%**, demonstrating its role in preserving previously correct grading decisions.

Network World Models as Environments for Algorithm Design on Complex Networks **Mar. 2026 – Present**
 Emory University, Machine Intelligence for Complex Systems Lab Prof. Liang Zhao, Atlanta, GA

- ◆ developed an **action-conditioned graph world model** that explicitly applies intervention semantics before learning subsequent diffusion dynamics. Integrated the model into an LLM-driven algorithm discovery loop, providing candidate evaluations and counterfactual feedback to guide program generation and refinement.
- ◆ Evaluated transition prediction, multi-step rollouts, and downstream decision quality on held-out graphs and unseen topologies. Achieved **90.2% candidate preference accuracy on held-out SBM graphs**; world-model-only candidate selection attained 1.01% regret, compared with 1.13% for simulator-based selection, using no trusted-simulator calls during candidate selection.

WORK EXPERIENCE (More Experiments Detailed in [Personal Webpage](#))

JPMorgan Chase & Co. — Volatility-Targeted Strategies Design **Jan. 2026 – Apr. 2026**
 Quantitative Research Intern Abhishek Namballa, New York, NY

- ◆ Co-developed an end-to-end volatility-targeting research pipeline for S&P 500 futures, covering data preprocessing, volatility forecasting, exposure allocation, and backtesting. Compared statistical and machine-learning estimators, including EWMA, GJR-GARCH, HAR-RV, and Random Forest, alongside regime-aware controllers and ensemble strategies.
- ◆ Evaluated estimator–controller combinations using forecasting accuracy, risk-adjusted returns, turnover, and drawdown. In the project backtests, HAR-RV with regime scaling achieved a Sharpe ratio of 0.602 versus 0.575 with naive scaling, while reducing maximum drawdown from 18.71% to 17.58% and increasing the Calmar ratio from 0.355 to 0.364.

SKILLS

- ◆ **Programming:** Python, SQL, Bash, MATLAB, Lean.
- ◆ **Tools & Platforms:** PyTorch, Hugging Face Transformers, verl, PyMARL, EPyMARL, MemOS, AgentDojo, NLIP, Qdrant, Git
- ◆ **Soft Skills:** Strong Self-motivation, independent research, interdisciplinary collaboration, analytical thinking, and effective multitasking.